

1. Administrative & Study Identification

Full Study Title: A Phase II, Randomized, Double-Blind, Placebo-Controlled, Multicenter Study to Evaluate the Efficacy, Safety, and Pharmacokinetics of Obinutuzumab in Adolescent Patients With Active Class III or IV Lupus Nephritis, Including an Evaluation of Open Label Safety and PK in a Cohort of Pediatric Patients (Aged 5 to < 12) **Registry Study Title:** A Study to Evaluate the Efficacy, Safety, and Pharmacokinetics of Obinutuzumab in Adolescents With Active Class III or IV Lupus Nephritis and the Safety and PK of Obinutuzumab in Pediatric Participants **Short Title/Acronym:** POSTERITY **Protocol Number:** WA42985 / 2021-000097-29 **NCT Number:** NCT05039619 **Posting ID:** 488634691547466382 **Sponsor / Company Name:** Roche (Hoffmann-La Roche) **Trial Status:** Recruiting **Investigational Product:** Obinutuzumab (Gazyva / Gazyvaro) **Phase of Development:** Phase II **Medical Monitor:** Head, EU/ROW Regulatory Affairs Pharmaceuticals Division, Hoffmann-La Roche

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy, safety, and pharmacokinetics (PK) of obinutuzumab compared to placebo in adolescent participants (aged 12 to <18 years) with biopsy-confirmed active Class III or IV proliferative lupus nephritis (LN). **Secondary Objectives:** To evaluate the open-label safety and PK of obinutuzumab in a younger pediatric participant cohort (aged 5 to <12 years) with LN.

2.2 Background & Rationale Systemic lupus erythematosus (SLE) is an autoimmune disease where autoantibodies attack the body's own tissues, leading to Lupus Nephritis (LN), a potentially life-threatening condition characterized by the irreversible loss of nephrons and kidney function. The current standard of care—a combination of corticosteroids and immunosuppressants (like mycophenolate mofetil)—remains limited, offering complete renal response in fewer than half of patients. Obinutuzumab is a type II anti-CD20 monoclonal antibody designed to deplete B-cells; while it has shown a positive effect in adult LN clinical trials, it needs rigorous evaluation for safety, PK, and efficacy in children and adolescents.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (for the adolescent cohort) **Blinding:** Double-

blind (adolescent cohort) / Open-label (pediatric cohort)
Randomization: Randomized (Adolescent cohort randomized 2:1 to obinutuzumab or placebo)

3.2 Planned Enrollment & Duration Target **\$N\$:** 40 participants (approx. 30 adolescents and 10 pediatric participants) **Number of Sites:** Multicenter (Sites across Europe, US, Canada, and Rest of World) **Estimated Study Start:** 2021 **Estimated Primary Completion:** June 2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 12 to <18 years at the time of randomization (or 5 to <12 years for the younger participant cohort once recruitment is open). 2. Biopsy-confirmed International Society of Nephrology and Renal Pathology Society (ISN/RPS) 2003 Class III or IV active LN demonstrated on renal biopsy performed in the 12 months prior to or during screening. Class V disease may be present in addition to Class III or IV. 3. Diagnosis of SLE according to the Systemic Lupus International Collaborating Clinics (SLICC) 2012 criteria. 4. Significant proteinuria defined by a UPCR > 0.5 based on a first-morning void (FMV) collection at screening. 5. During the 12 months prior to or during screening, received at least one dose of pulse-range IV methylprednisolone (typically 30 mg/kg, max 1000 mg/dose) or equivalent for the treatment of the current episode of active LN.

4.2 Exclusion Criteria 1. Severe, active central nervous system (CNS) SLE, including retinitis, poorly controlled seizure disorder, acute confusional state, myelitis, stroke, cerebellar ataxia, or dementia. 2. Sclerosis in >50% of glomeruli on renal biopsy. 3. Purely chronic Class III(c) or Class IV(c) disease on renal biopsy, defined as the absence of any active lesions. 4. Presence of rapidly progressive glomerulonephritis. 5. Participants with isolated pure Class V Lupus Nephritis disease. 6. Intolerance or contraindication to study therapies, prior treatment with obinutuzumab, or known hypersensitivity. 7. Active infection of any kind (excluding fungal infection of nail beds) or any major episode of infection requiring hospitalization or treatment with IV anti-infective medications within 4 weeks prior to screening, or completion of oral anti-infectives within 2 weeks prior to randomization. 8. History of or currently active primary or secondary immunodeficiency, including known history of HIV infection and other severe immunodeficiency blood disorders, or history of serious recurrent or chronic infection. 9. History of or current cancer, including solid tumors, hematological malignancies, and carcinoma in situ (except basal cell carcinoma and squamous cell carcinoma of the skin that have been excised and cured) within the past 5 years. 10. Significant or uncontrolled concomitant medical disease precluding participant participation. 11. Currently active alcohol or drug abuse or history of alcohol or drug abuse.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Obinutuzumab 1000 mg for participants with a body weight of ≥ 45 kg. Participants weighing < 45 kg will receive a weight-adjusted dose of 20 mg/kg. Administered on Day 1, Day 14, Week 24, Week 26, and Week 52. **Route of Administration:** Intravenous (IV) infusion. **Duration of Treatment:** 52 weeks of active dosing, with total clinical trial observation lasting up to 2.5 to 3.5 years.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care therapies for LN, including Mycophenolate Mofetil (MMF), systemic corticosteroids, and permitted symptomatic medications including paracetamol (ATC Code: N02BE01). **Prohibited:** Other investigational biological agents or concurrent anti-CD20 therapies (e.g., rituximab).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, SLE/LN Diagnosis Review, Renal Biopsy Assessment, Proteinuria Labs
Baseline	Day 1	Randomization, First IV Infusion (Obinutuzumab or Placebo), Baseline PK/Safety Labs
Interim Dosing	Day 14, Weeks 24, 26, 52	IV Infusions, PK/PD Sampling, B-cell monitoring, Efficacy & Safety Assessments
Follow-up	Post-Week 52	Long-term Safety Monitoring, Renal Response Assessments (up to 2.5 - 3.5 years)
End of Study	Year 2.5 - 3.5	Final Renal Function Assessment, Exit Interview, Safety Labs

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Percentage of Participants who Achieve a Complete Renal Response (CRR)** in Adolescent Participants (AP). CRR is defined as achievement of *all* of the following: * Urinary protein-to-creatinine ratio (UPCR) < 0.5 g/g * Estimated Glomerular Filtration Rate (eGFR) $\geq 85\%$ of baseline * No occurrence of intercurrent events 2. **Percentage of Participants with Adverse Events** in Pediatric Participants (PP).

7.2 Secondary & Exploratory Endpoints 1. Percentage of Participants Achieving a CRR (AP). 2. Percentage of Participants who Achieve CRR with Successful Prednisone Taper (AP). 3. Percentage of Participants who Achieve a Partial Renal Response (PRR) (AP). 4. Pharmacodynamics (PD), specifically CD19+ B-cell depletion and reconstitution over time. 5. Overall safety profile (incidence and severity of AEs, SAEs, and Infusion-Related Reactions).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of AEs with specific focus on Infusion-Related Reactions (IRRs) and neutropenia. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness, particularly severe infections due to immunosuppression. **Data Monitoring Committee (DMC):** Independent external DMC to evaluate unblinded safety and PK data, particularly prior to opening recruitment for the younger pediatric cohort (5 to <12 years).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy analysis; Safety Set for all participants receiving at least one dose of study medication. **Sample Size Justification:** Target $N=40$ (approx. 30 adolescents randomized 2:1 and 10 pediatric participants) is deemed sufficient for a Phase II evaluation of preliminary efficacy, safety, and PK profiles in a rare pediatric/adolescent LN population. **Handling of Missing Data:** Non-responder imputation (NRI) for primary renal response endpoints; Multiple Imputation (MI) or Mixed Models for Repeated Measures (MMRM) for continuous variables.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence, correct IV infusion protocols, and accurate pediatric weight-based dose adjustments. **Audit Trail:** Validated eCRF systems with comprehensive audit trails, utilizing query management for discrepancy resolution.

11. Study Identifiers & Governance

Primary ID: WA42985 **Registry Number:** NCT05039619 **Sponsor/Lead Organization:** Roche (Hoffmann-La Roche) **Study Title:** POSTERITY **Disease Areas:** Lupus Nephritis (Preferred UMLS Name: Lupus Nephritis)

12. Clinical Framework (Lupus Nephritis Specific)

Primary Condition: Active Proliferative Lupus Nephritis (LN) **Diagnosis Threshold:** ISN/RPS 2003 Class III or IV on Renal Biopsy **Medical Methodology:** B-cell depletion via anti-CD20 monoclonal antibody combined with Standard of Care (MMF + Corticosteroids)

Optimization Target: Complete Renal Response (CRR) and sustained CD19+ B-cell depletion

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for severe autoimmune nephritis + Investigational IV Infusion **Education Protocol (HCP):** Training on weight-based dose calculations (\$20\$ mg/kg for \$< 45\$ kg vs \$1000\$ mg flat dose) and IRR management. **Education Protocol (Patient):** Patient/Guardian education on prolonged immunosuppression risks, strict adherence to background MMF/steroids, and infection prevention. **Quality Audit Mechanism:** Centralized laboratory tracking of B-cell counts and PK exposures to ensure adequate drug targeting.

14. Observation & Follow-Up Schedule

Total Duration: Up to 2.5 to 3.5 years **Onsite Visit Cadence:** Day 1, Day 14, Weeks 24, 26, 52 (Dosing visits), with subsequent follow-up protocols. **Remote Monitoring Frequency:** Routine telemedicine checks for safety and symptom flare-ups between onsite infusion visits. **Checklist Review Interval:** Regular interval reviews of proteinuria (UPCR) and serum creatinine to monitor renal response status.

15. Sign-off & Approval

Role	Name	Signature	Date		:---		:---		:---		:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]									
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]									
Statistician	[Name]	_____	[DD-MMM-YYYY]									